Volume Vi Algebra Reference

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Introduction to Algebraic Structures

0.1 What Is a Mathematical Structure?

0.1.1 Algebraic Structures

Definition 0.1.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Definition 0.1.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Definition 0.1.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

0.1.2 Laws of Composition

Definition 0.1.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Definition 0.1.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Definition 0.1.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition 0.1.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

0.1.3 Distinguished Elements

Definition 0.1.8 (Left Identity)**.** An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 0.1.9 (Right Identity)**.** An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 0.1.10 (Identity / Neutral Element)**.** An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Proposition 0.1.11 (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique. Go to proof.*

Definition 0.1.12 (Left Absorbing Element)**.** An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 0.1.13 (Right Absorbing Element)**.** An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 0.1.14 (Absorbing Element)**.** An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Proposition 0.1.15 (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique. Go to proof.*

0.1.4 Inverses

Definition 0.1.16 (Left Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 0.1.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 0.1.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Definition 0.1.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Proposition 0.1.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique. Go to proof.*

Definition 0.1.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 0.1.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 0.1.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group. Go to proof.*

0.1.5 Properties of Laws

Definition 0.1.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Theorem 0.1.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization. Go to proof.*

Definition 0.1.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 0.1.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 0.1.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Definition 0.1.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

0.1.6 Regularity Properties

Definition 0.1.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.1.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.1.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 0.1.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.* *Go to proof.*

Definition 0.1.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

0.1.7 One Internal Law

Definition 0.1.37 (Magma)**.** A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 0.1.38 (Semigroup)**.** A **semigroup** is a magma whose operation is associative.

Definition 0.1.39 (Monoid)**.** A **monoid** is a semigroup with an identity element.

Definition 0.1.40 (Group)**.** A **group** is a monoid in which every element has an inverse.

Definition 0.1.41 (Abelian Group)**.** An **abelian group** is a group whose operation is commutative.

Definition 0.1.42 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 0.1.43 (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.* *Go to proof.*

Proposition 0.1.44 (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.* *Go to proof.*

Proposition 0.1.45 (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 0.1.46 (Socks–Shoes)**.** *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

0.1.8 Two Internal Laws

Definition 0.1.47 (Rng)**.** A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 0.1.48 (Ring)**.** A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 0.1.49 (Commutative Ring)**.** A **commutative ring** is a ring whose multiplication is commutative.

Definition 0.1.50 (Integral Domain)**.** An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 0.1.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 0.1.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 0.1.53 (Multiplication by Zero)**.** *In a ring,* $a \cdot 0 = 0 \cdot a = 0$ *for all* a*.* *Go to proof.*

Proposition 0.1.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$*.* *Go to proof.*

Proposition 0.1.55 (Cancellation in Integral Domains)**.** *In an integral domain, if* $a \neq 0$ *and* $ab = ac$*, then* $b = c$*.* *Go to proof.*

Proposition 0.1.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain.* *Go to proof.*

Proposition 0.1.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field.* *Go to proof.*

0.1.9 The Structural Lattice

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Basic Definitions and Theorems

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Algebraic Geometry

0.16 Algebraic Geometry

0.16.1 Preliminary Definitions

Definition 0.16.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Definition 0.16.2 (Commutative Ring)**.** A ring R is *commutative* if
$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Polynomial Rings

Definition 0.16.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression
$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$
where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Definition 0.16.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is
$$\deg(f) := n.$$
The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Definition 0.16.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:
$$\left(\sum_i a_i x^i \right) + \left(\sum_i b_i x^i \right) := \sum_i (a_i + b_i) x^i,$$
$$\left(\sum_i a_i x^i \right) \cdot \left(\sum_j b_j x^j \right) := \sum_k \left(\sum_{i+j=k} a_i b_j \right) x^k.$$

Polynomial Rings in Several Variables

Definition 0.16.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \ldots, x_n],$$

is defined inductively by

$$R[x_1, \ldots, x_n] := R[x_1, \ldots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_{\alpha} a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Definition 0.16.7 (Monomial)**.** A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Definition 0.16.8 (Total Degree)**.** The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Ideals

Definition 0.16.9 (Ideal)**.** Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Definition 0.16.10 (Generated Ideal)**.** Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by $f_1, \ldots, f_m$* is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated.*

Definition 0.16.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \ldots, f_m \in R$ such that $I = \langle f_1, \ldots, f_m \rangle$.

Definition 0.16.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Theorem 0.16.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*
In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k. *Go to proof.*

Quotient Rings

Definition 0.16.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Proofs

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